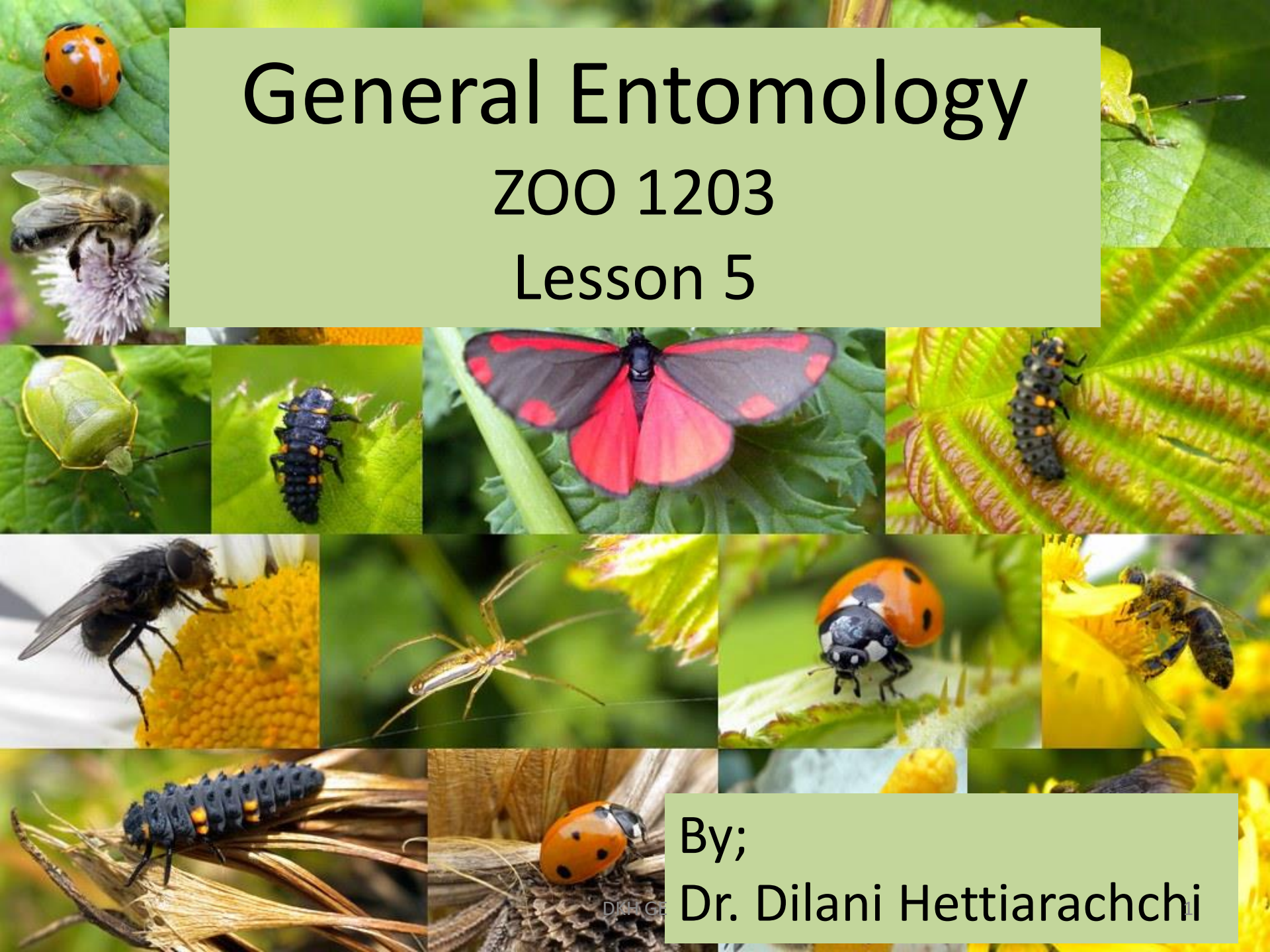


General Entomology

ZOO 1203

Lesson 5



By;

Dr. Dilani Hettiarachchi

Lesson Objectives

After completion you should be able to;

- Explain the structure of thorax of a general insects
- Identify different leg patterns found in insects and discuss the adaptations of leg types
- Explain the wing adaptations of insects
- Describe the appendages found in the abdominal region of insects

Thorax

- The thorax;
 - Prothorax – top is called pronotum
 - Mesothorax
 - Metathorax
- Wings and legs are attached to the thoracic region

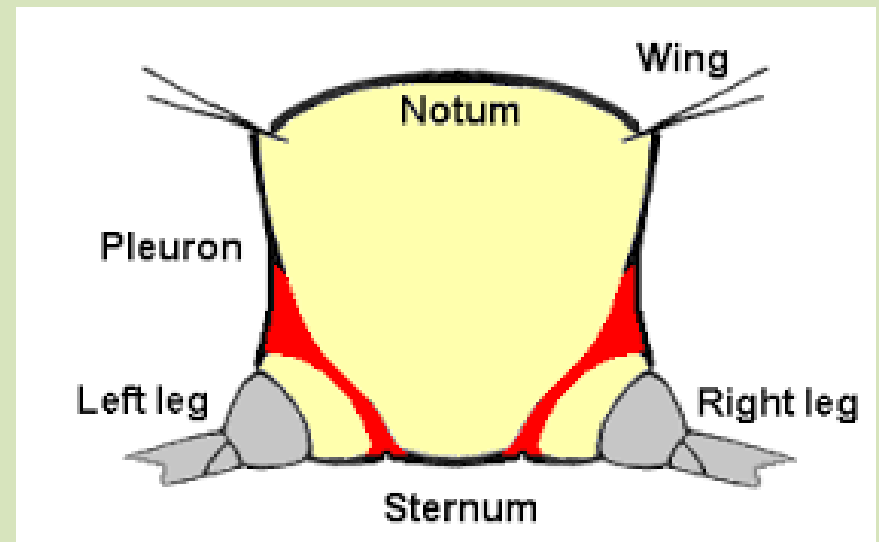


- Primitive wingless (apterigote) insects and immature insects the three thoracic segments are similar in size and structural complexity
- Mesothorax and metathorax are enlarged relative to prothorax in winged insects
- They bear the musculature for wing attachment
- Spiracles are present laterally on the 2nd and 3rd thoracic segment - at most one pair per segment

- Each segment is composed of hardened plates that gives it rigidity
- They are:
 - dorsal tergum/notum (pronotum, mesonotum, and metanotum)
 - pair of bilateral pleura
 - ventral sternum (prosternum, mesosternum, and metasternum)

Nota

- Wing bearing segments
- Vary in number, nature
- further subdivided into an anterior **scutum** and a posterior **scutellum**



Pair of bilateral pleura

- Prominent in winged insects
- In wingless forms this is composed of a group of subcoxal sclerites

Sterna

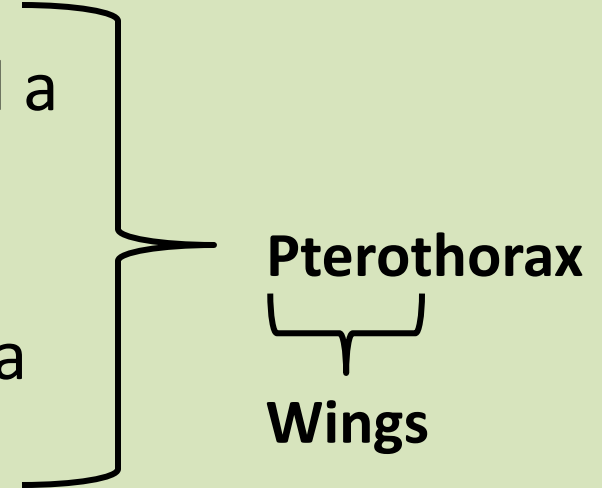
- The degree of ventral sclerotization on the thorax varies greatly in different insects
- Sterna is the ventral sclerite bearing site

- Appendages of thorax
 - Three pairs of legs
 - Two pairs of wings

❑ Prothorax- a pair of Pro legs

❑ Mesothorax- A pair of Meso legs and a pair of fore wing

❑ Metathorax- A pair of meso legs and a pair of hind wings



Class Insecta

Wingless

Winged or secondarily
wingless

Apterygota

Pterygota

Eg: silverfish

jumping bristletails

Orders: Archaeognatha

Cercopoda

Monura

Zygentoma



Wings

- Insects are the only winged invertebrate
- May bear a single pair or two pairs

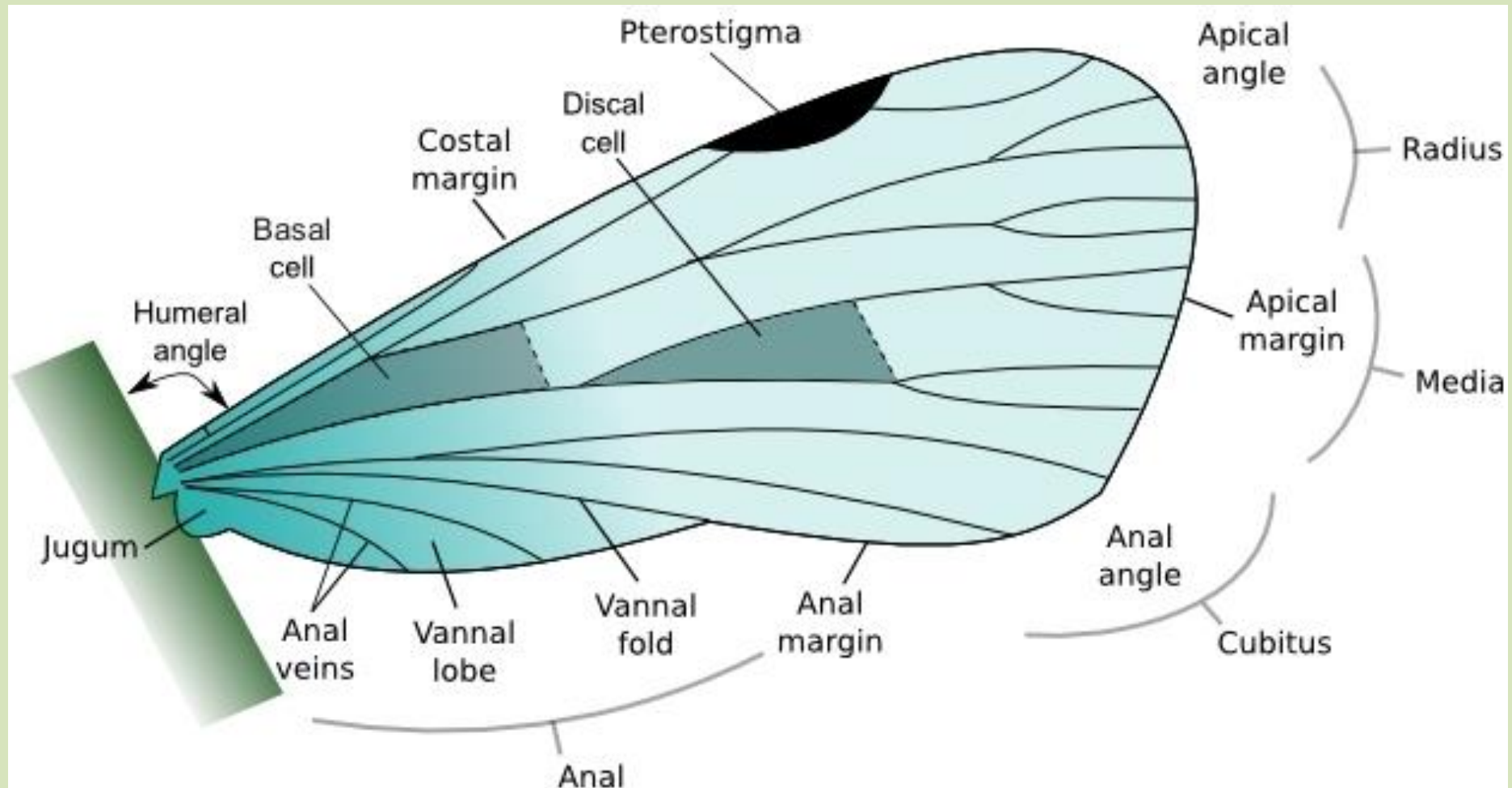
Size

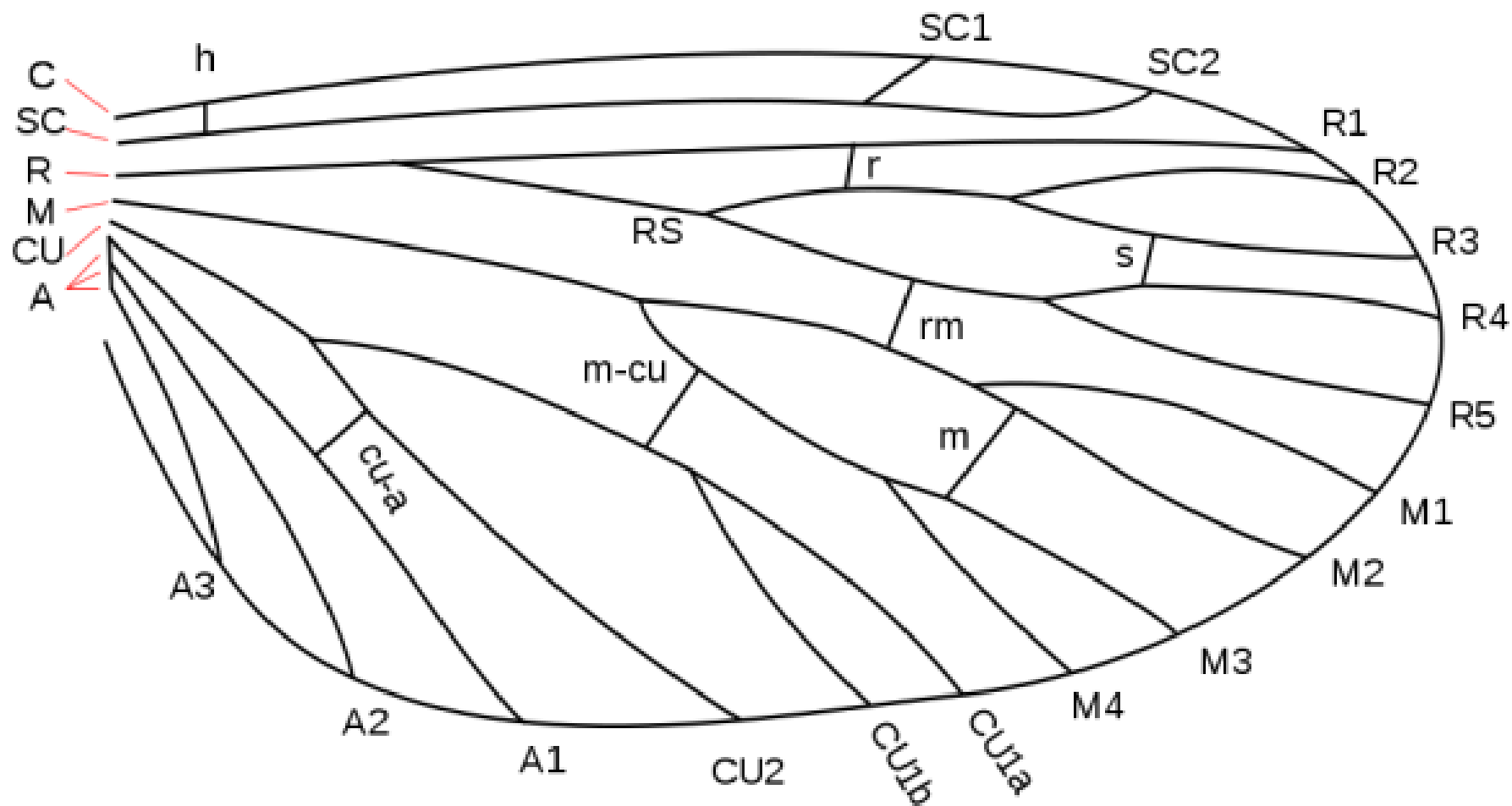
- Quite large or extremely small
- The Atlas moth of Australia has been said to have a wingspan of 14 inches
- In many hindwing is smaller than the forewing

Venation

- Tremendous variations
- Thus use as a taxonomic character
- Extensively reduced simplified wing venations (wasps)
- Highly complex network of wing venation (dragonfly)

- Veins also vary in thickness (cicads – very thick; scorpionfly – very thin)
- cuticle surrounding the veins becomes thickened and more heavily sclerotized to provide strength and rigidity to the wing





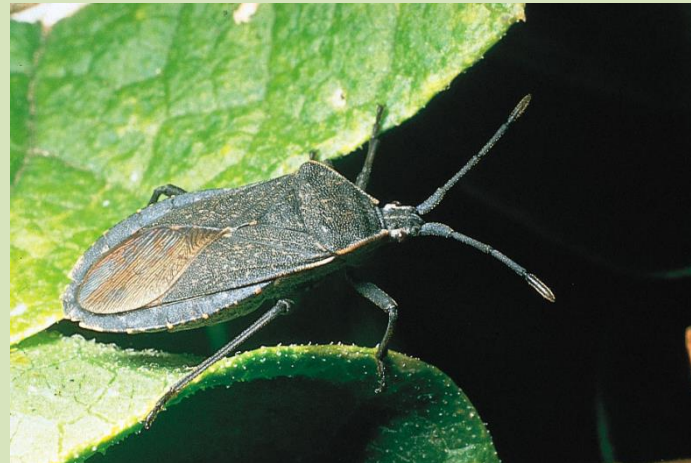
WING VENATION

Function and texture

- Function → flight
- However in some cases they are used for other purposes:
 - form a protective armor (elytra) – in beetles
 - production of sound – Field cricket: Orthoptera
 - stability of flight halteres – hind wing modification in housefly
 - use as a fan – honey bee: as a community



- Wings are named:
 - Elytra – forewings hard (Coleoptera)
 - Hemelytra – half elytra (Hemiptera)
 - Tegmen – parchment like (orthoptera)



Resting position

- when not been used the wings held in various position
- Members of primitive orders are unable to fold the wings over the abdomen
- They hold them vertically (mayfly and damselfly) or horizontal (dragonfly) – group Paleoptera



- Other insects are able to flex the wings over the abdomen at rest – Neoptera group



- Many homoptera and others (neuroptera) hold the wing roof like over the abdomen – Neoptera group



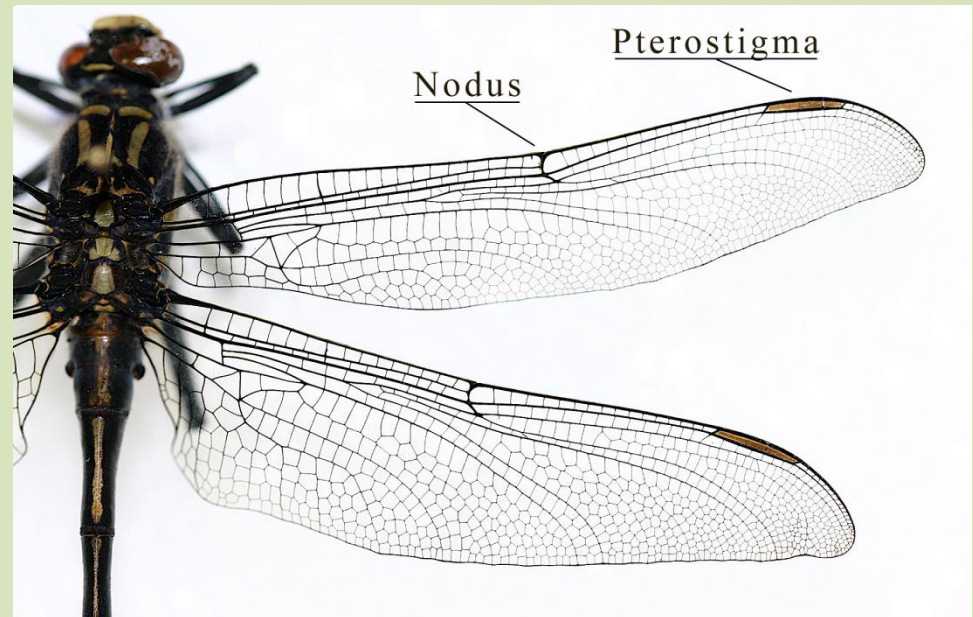
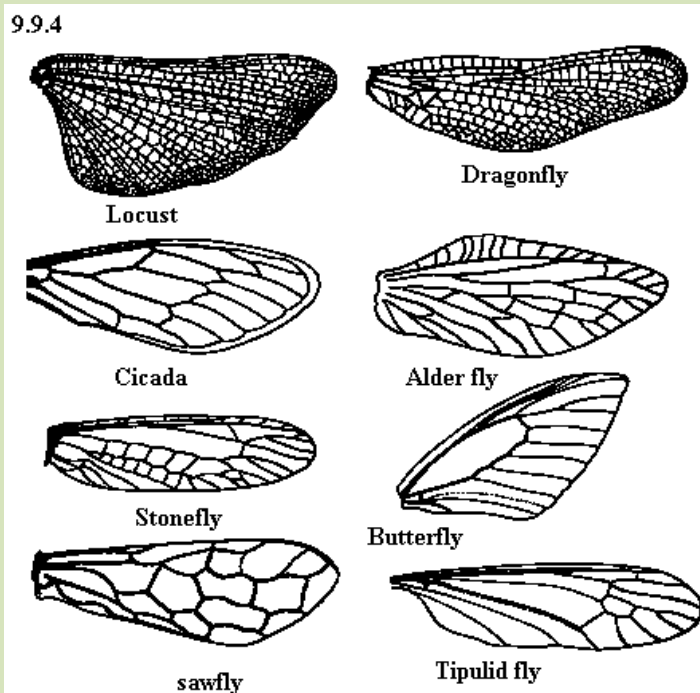
- Neoptera group can fold their wings in many patterns
- They can fold the wing flat above the abdomen
- It protect the wing from physical damage
- As it's folded compactly it allow the insect to crawl into tight spaces
- Wing folding make use of complex mechanical adaptation
 - Fold lines
 - Wing veins of variable diameter, thickness and shape

Coloration

- Many have clear/hyaline/opaque color unpigmented wings
- In some wing coloration is well developed
- Coloration maybe due to pigments within the integument itself or by covering of minute pigmented scale
- Or physically resemble thin layers or diffraction gratings in their effect upon impinging light
- Has a protective function



- Scales of butterflies and moths are modified setae
- Wings of caddisflies are covered with tiny hairs (modified setae)
- Hairs of two types may occur on the wings:
 - microtrichia - small and irregularly scattered
 - macrotrichia - larger, socketed (may be restricted to veins)



Legs

- Generalized insect leg consists of six segments:

1. A basal Coxa

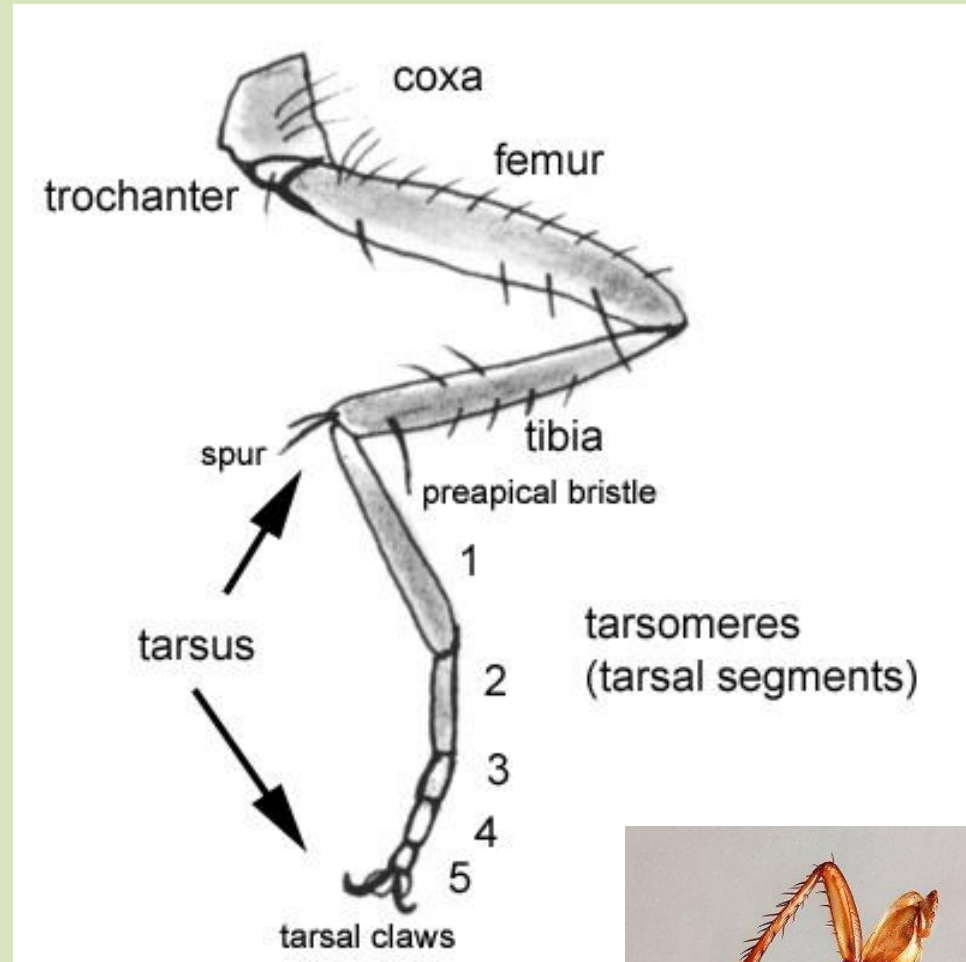
2. small trochanter

3. A femur

4. A tibia

5. A segmented tarsus

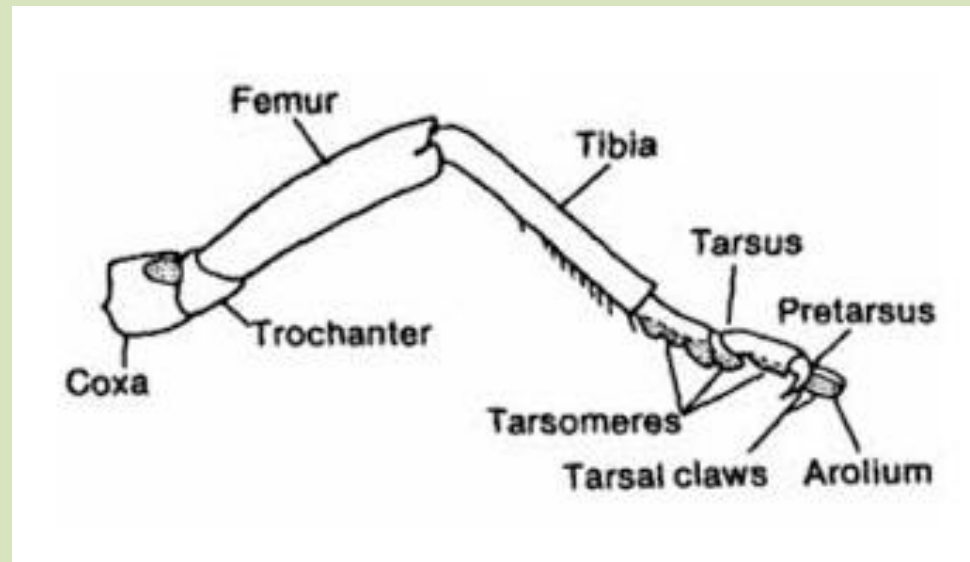
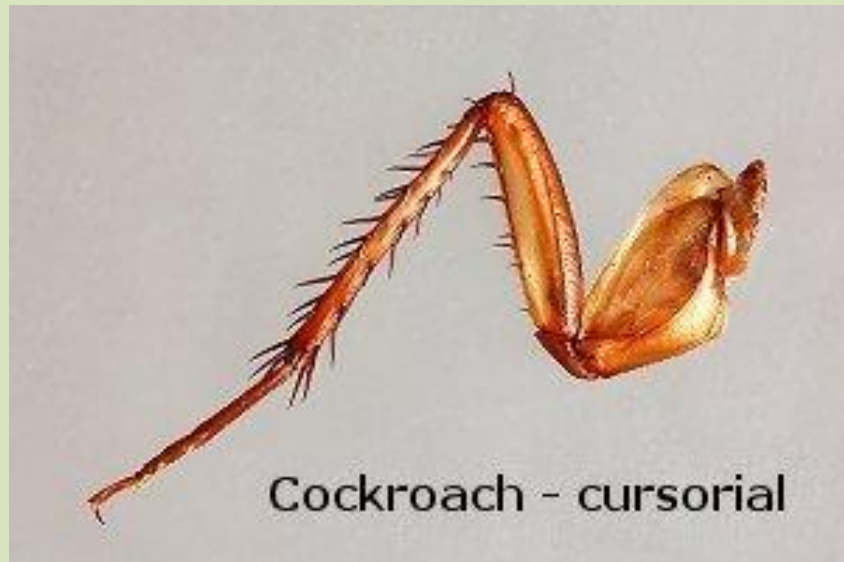
6. A pretarsus/ claws



- Tibia and femur are the longest
- Tarsus is divided into tarsomeres
- The pretarsus may consists of a single claw; usually with a pair
- Many modifications to the mode of life of insect

1. Cussorial (running)

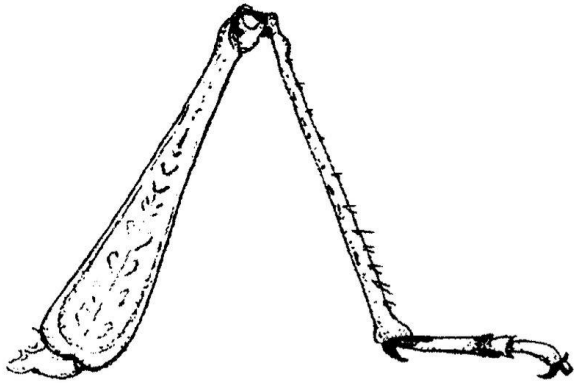
- Have a well developed femur and tibia on all legs



2. Saltatorial (jumping)

- Disproportionately developed hind femur and tibia
- Femur is not swollen
- Eg: hind legs of Grasshoppers (Orthoptera)

**Saltatorial - (Saltatorial =
Leaping ; Jumping Leg)**



Grasshopper - saltatorial



3. Natatorial (swimming)

- Tibia and/or tarsi of one or more pairs of legs usually are modified for swimming with fringes of long slender hairs
- Eg: aquatic beetles (Coleoptera) and bugs (Hemiptera)



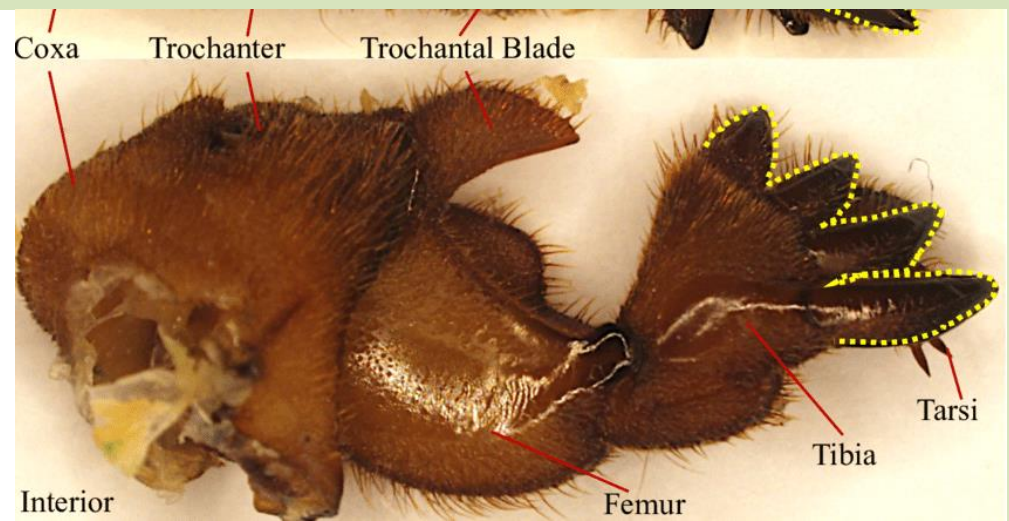
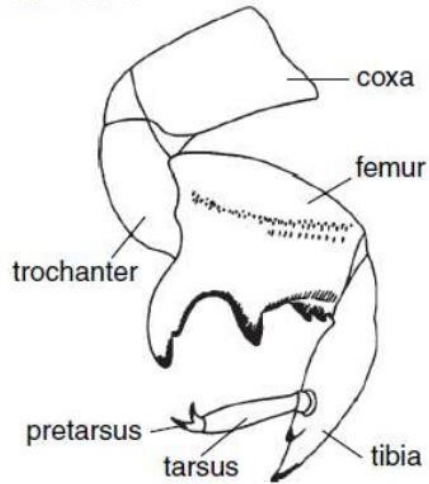
Backswimmer - natatorial



4. Fossorial (burrowing/ ground dwelling)

- Tibia of the forelegs enlarged and modified for digging
- Eg: fore legs of mole cricket (Orthoptera: Gryllotalpidae), scarab beetles (Scarabaeidae)

(a) digging

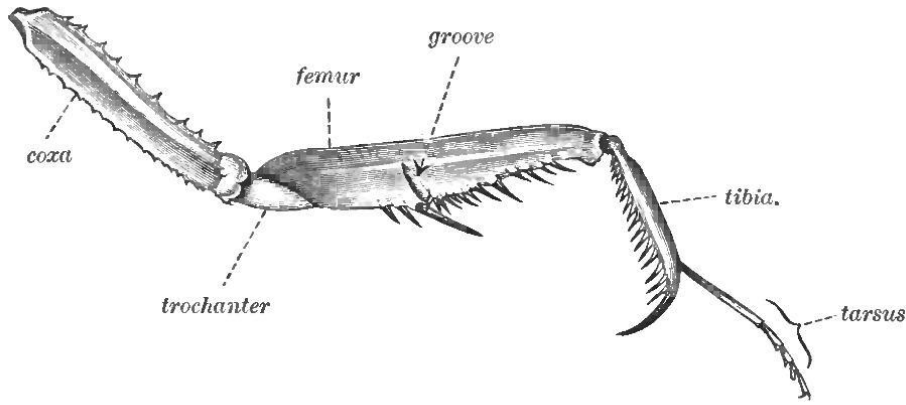


Mole Cricket



5. Raptorial (praying mantis)

- Femur and tibia are heavily built and femur with serrated margins
- Fore legs of praying mantis

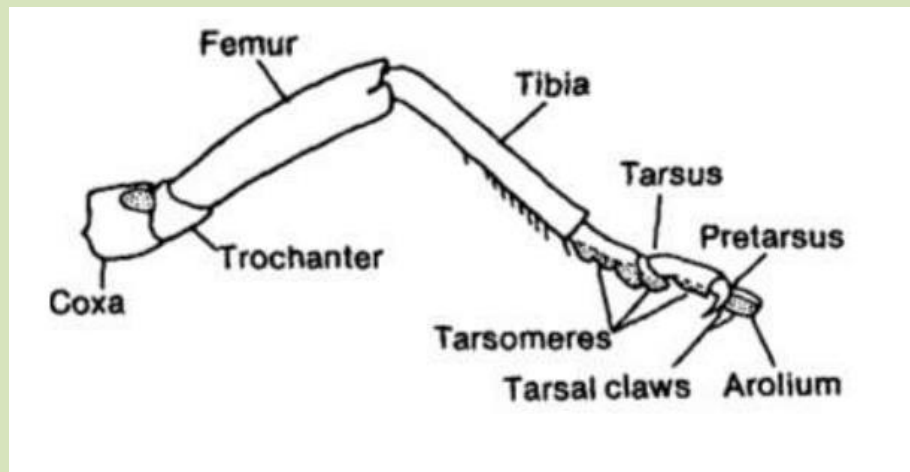


Leg of a Mantis.



6. Ambulatorial (walking)

- Femur and tibia are long and adapted for walking
- Foreleg and middle leg of grasshoppers and cockroach

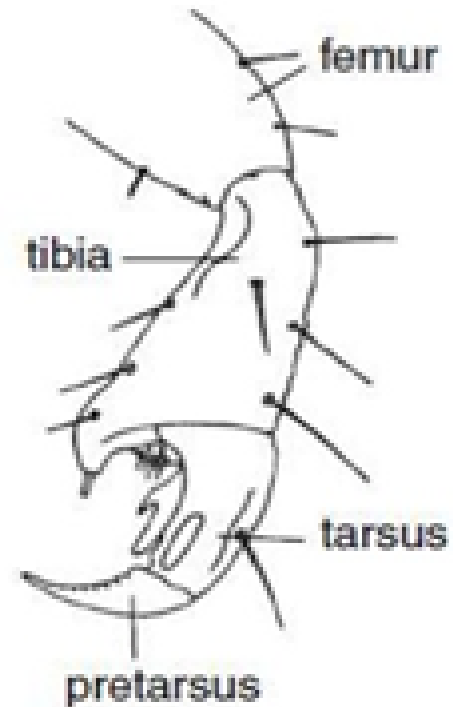


7. Scansorial (climbing)

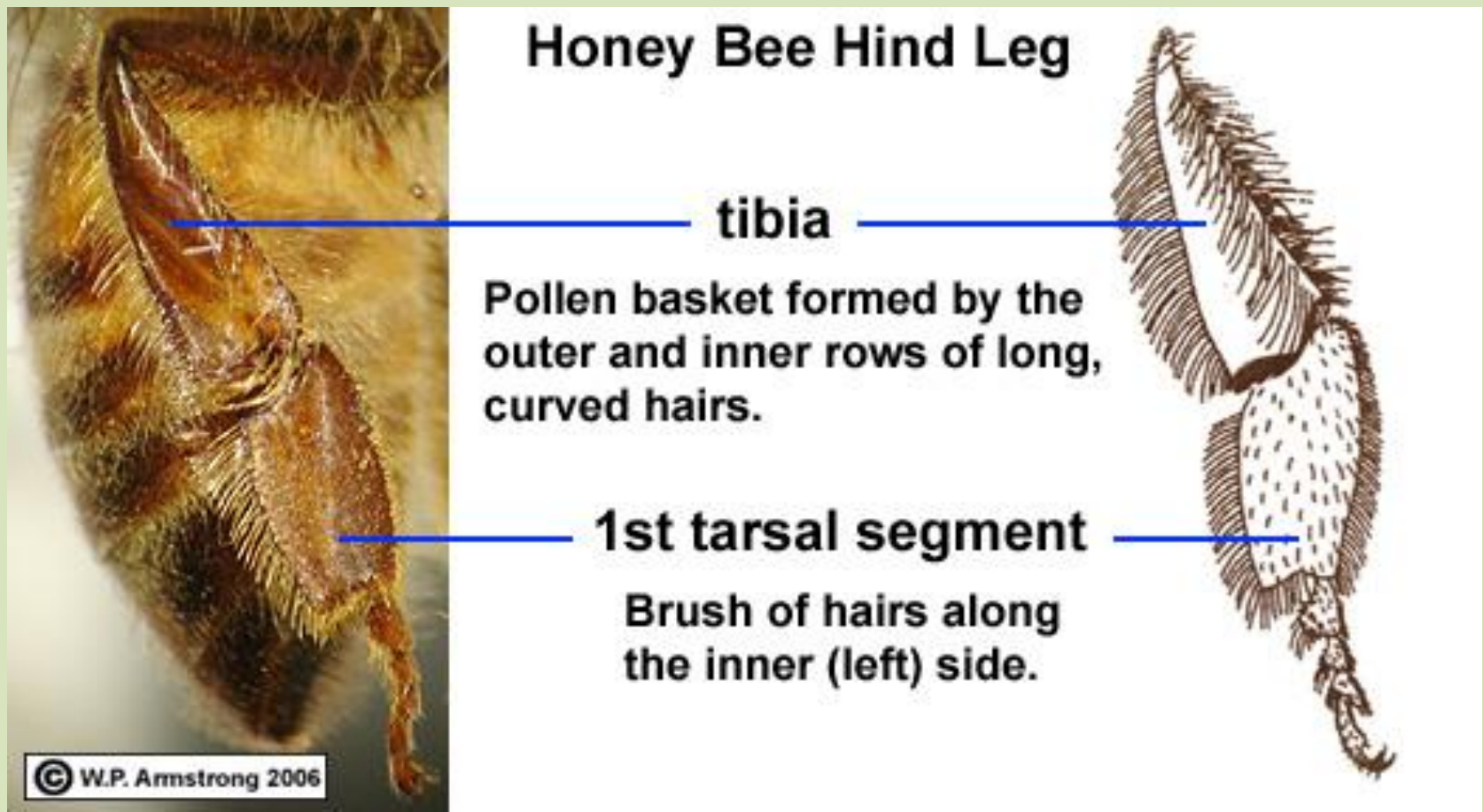
- Tibia is stout and at the end have a thumb like process with a distal tarsal segment
- Curved pre-tarsal claw is present



(b) grasping



- Apart from these named modifications there are few others like in the case of honey bees
- They have a pollen basket in their hind leg and modified to carry pollen



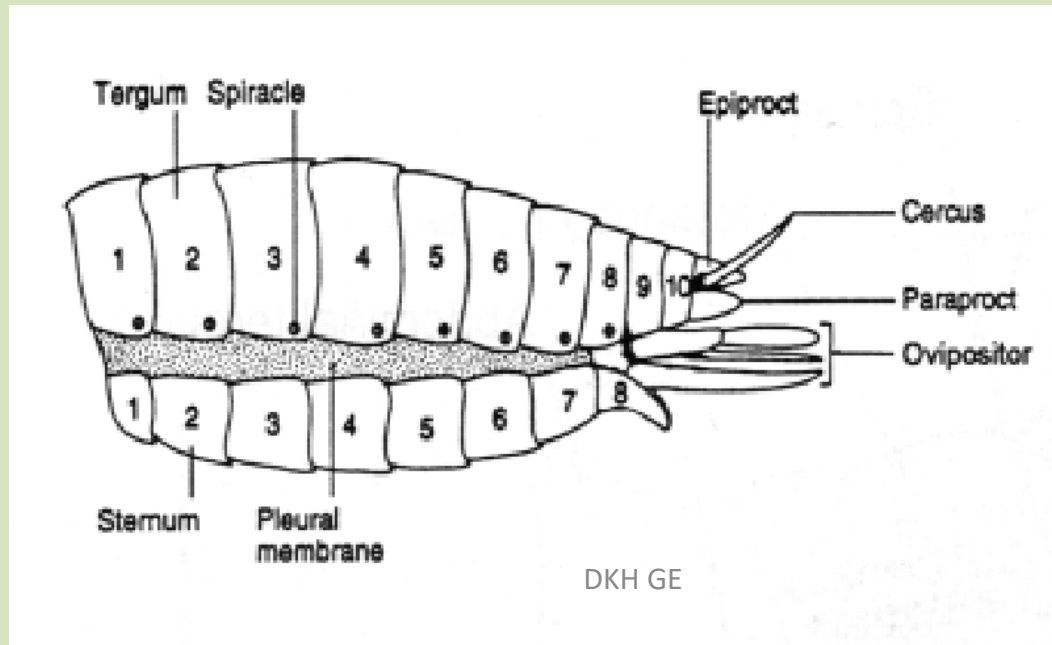
Abdomen

- In insect evolution, tendency is to reduce the no. of segments
- Varying number of legless segments
- Primitive no. of true metameres are 11 and the terminal segments which bears anus
- However , this number could vary from insects to insects
- Terminal segment is called periproct/telson
- It bears the anus
- In hymenoptera 1st segment is constricted to form a pedicel



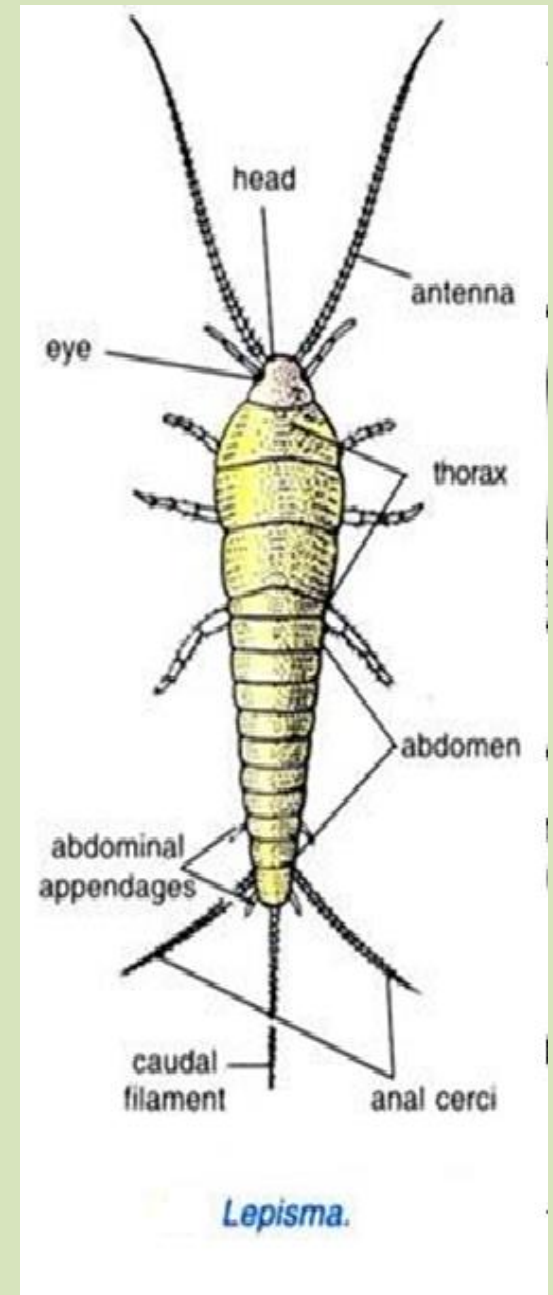
Non-Genital abdominal appendages

- There are three plates in the terminal segment;
 - Epiproct – dorsal to the anus
 - Two paraprocts – either side of the anus
- Spiracles are typically found on either side of the first eight abdominal segments



- Terminal segment may bear a pair of appendages
 - the cerci
- In some insects, the last abdominal segment bears a long , median filament
- Cerci vary in from: forcep-like (earwig), antennae-like (crickets), reduced (cockroachers), absent (bugs)
-
- In certain insects they are very long.
- They are believed to be tactile in some, olfactory in others and in some males they aid in holding the female in copulation

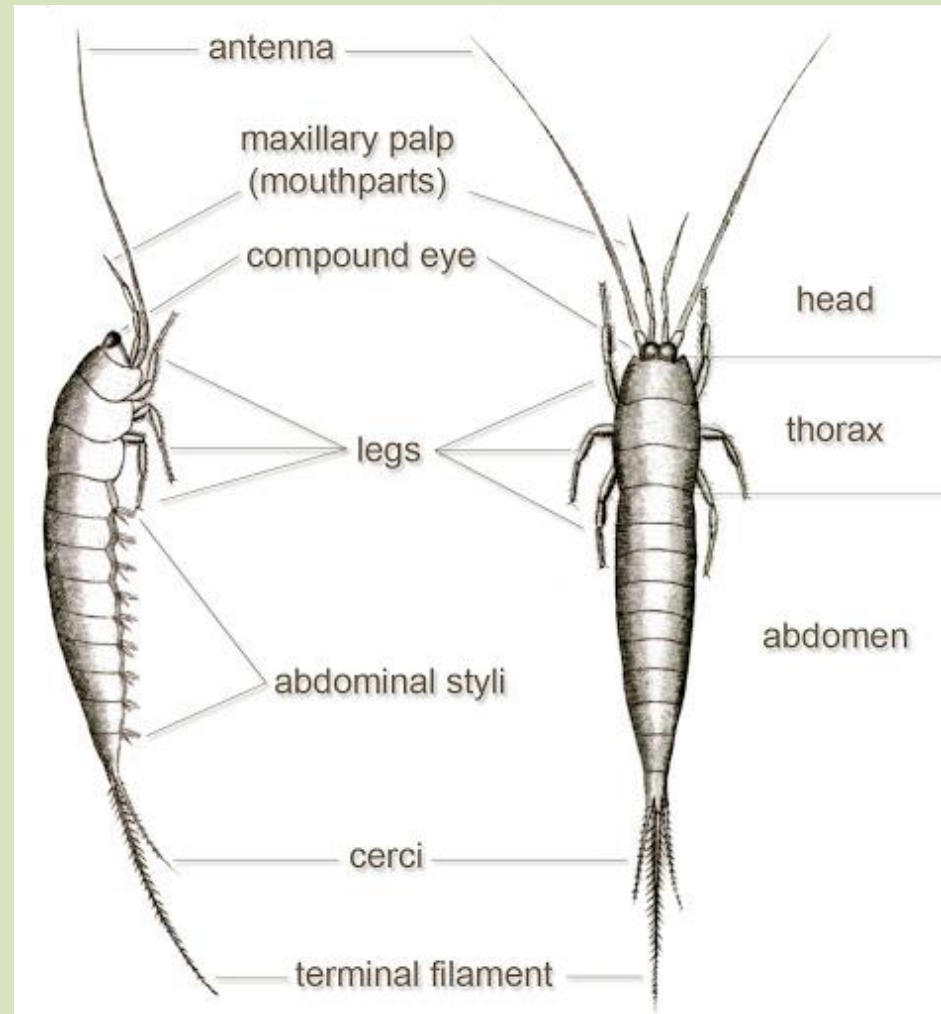




– the styli

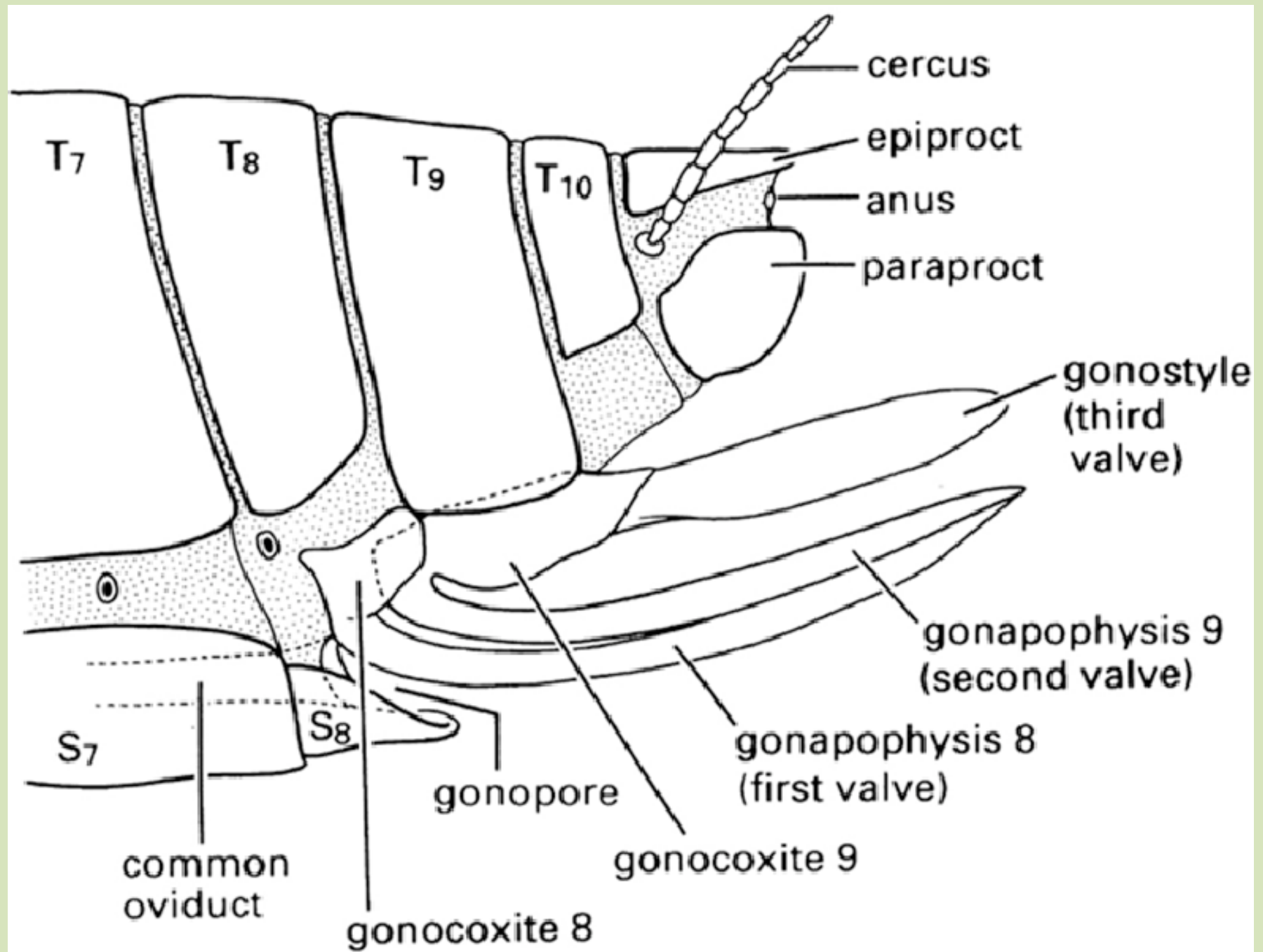
- Small, rudimentary, leg-like structure
- Extend from the abdomen of some adult insects

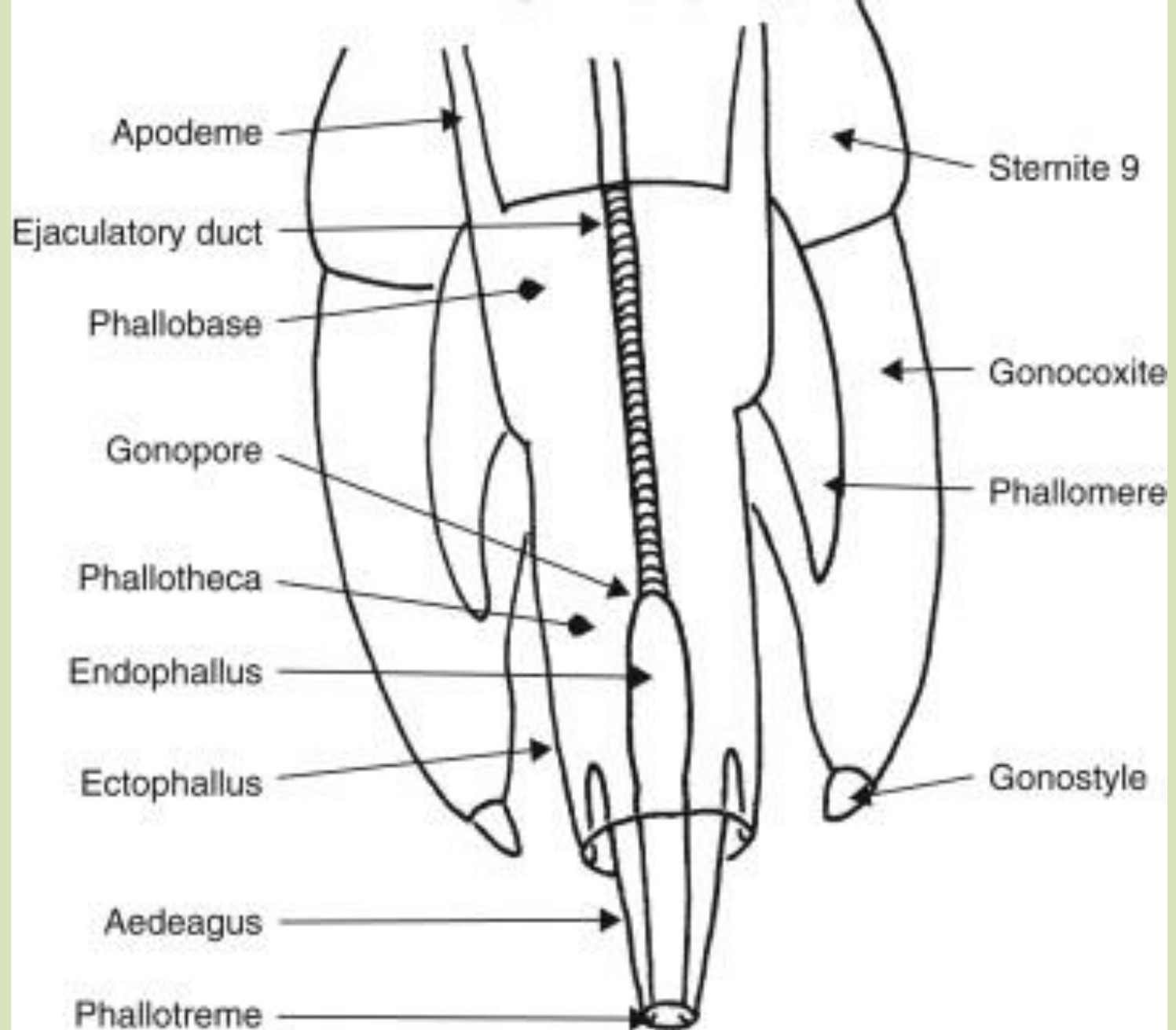
Eg: silverfish
(Order: Zygentoma)



Genital abdominal appendages

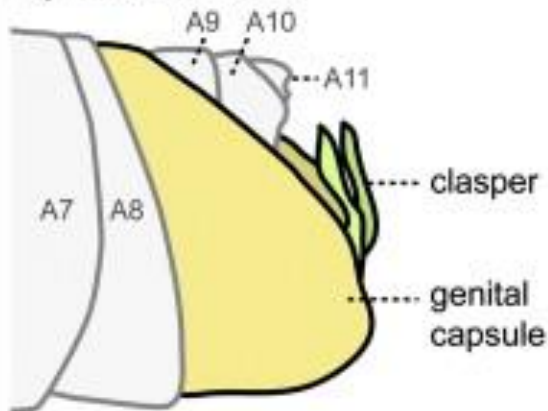
- External genitalia are;
 - Ovipositor – females
 - copulatory apparatus – males
- By modifying 8th and 9th abdominal segments – ovipositor (egg laying)
 - composed of pairs of valvifers – 1 pair on 8th and 2 pairs on 9th
- The female gonopore (reproductive opening) - on or posterior to the 8th or 9th segment
- Male external copulatory apparatus (penis or aedeagus) – 9th abdominal segment



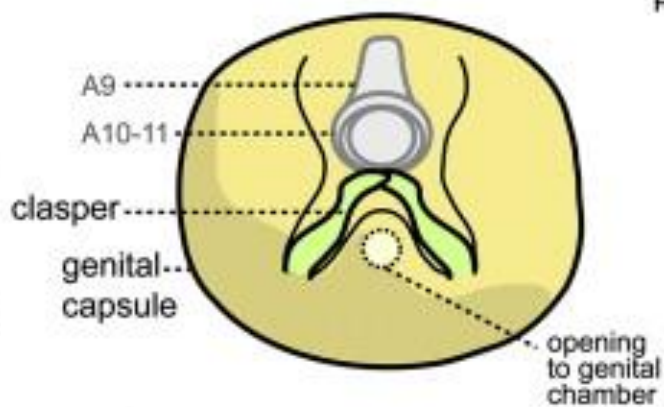


Oncopeltus fasciatus

A) male lateral



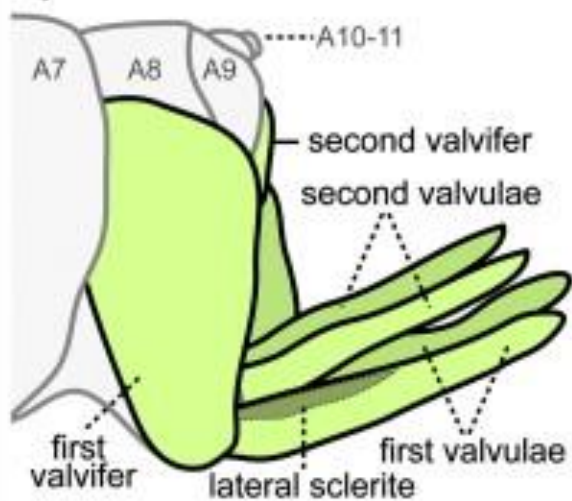
B) male posterior



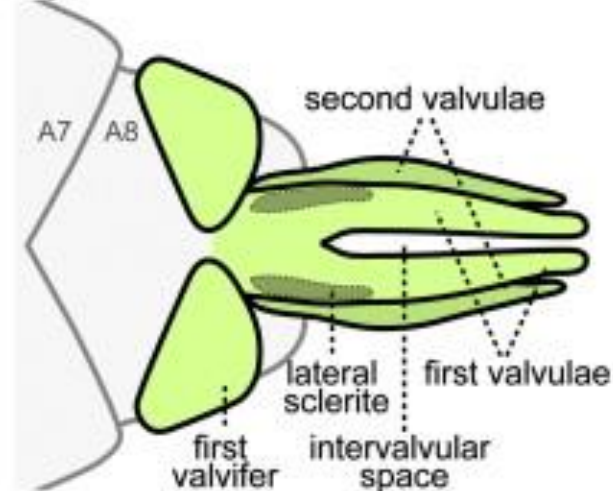
C) male copulatory organ



D) female lateral

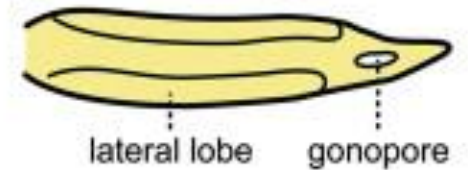


E) female ventral

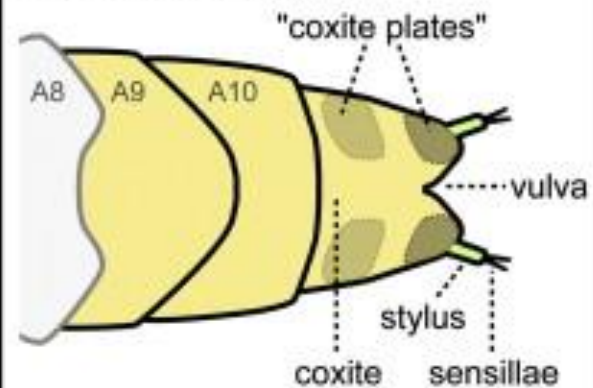


Tribolium castaneum

F) male aedeagus



G) female ventral



Modification of the general body form

- Shape of many insects are adaptations to a given environment



THANK YOU!

